CHERN CLASSES OF A TENSOR PRODUCT OF FINITE LOCALLY FREE MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's closing sentence at source lines 8631-8632, 'The result follows from a formal computation which we omit.', is retained verbatim at the end of the counted proof. The reduction to invertible successive quotients, the lexicographic filtration of the tensor product and the three product expressions for the total Chern classes are all inside the span; only the expansion of those products into the displayed coefficients is left to the reader, which is the routine residue LANE_BRIEF section 5 accepts.

Allowed standard prerequisites (editorial).

- Reduction to invertible modules by a filtration, as in the dual case. Same chapter chow.tex, label lemma-chern-classes-dual, invoked at proof line 8588. The same argument as in that lemma lets one assume the two modules are successive extensions of invertible modules, so that the total Chern classes are products of linear factors.
- The total Chern class of a module filtered by invertible modules is the product of the linear factors. Same chapter chow.tex, label lemma-chern-filter-by-linebundles, invoked at proof line 8622. Turns the two filtrations and the induced filtration of the tensor product into the three product expressions for the total Chern classes of the two modules and of their tensor product.

Source reference. Original lemma 0FA5: sources/stacks-chow/pinned-original-source.tex, source lines 8520–8520. Navigation pointer for \ref{lemma-chern-classes-dual} used

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inside the counted span (line 8588); the argument that consumes it is retained in the span.

Source reference. Original lemma 02UJ: sources/stacks-chow/pinned-original-source.tex, source lines 8022–8022. Navigation pointer for \ref{lemma-chern-filter-by-linebundles} used inside the counted span (line 8622); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 8568); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 02QL let X be locally of finite type over S. Let $\mathcal{E}$ and $\mathcal{F}$ be a finite locally free $\mathcal{O}_X$-modules of ranks r and s. Then we have*

$$c_1(\mathcal{E} \otimes \mathcal{F}) = rc_1(\mathcal{F}) + sc_1(\mathcal{E})$$

$$c_2(\mathcal{E} \otimes \mathcal{F}) = rc_2(\mathcal{F}) + sc_2(\mathcal{E}) + \binom{r}{2} c_1(\mathcal{F})^2 + (rs-1)c_1(\mathcal{F})c_1(\mathcal{E}) + \binom{s}{2} c_1(\mathcal{E})^2$$

and so on in $A^*(X)$.

Proof. Arguing exactly as in the proof of Lemma 0FA5 we may assume we have invertible $\mathcal{O}_X$-modules $\mathcal{L}_i$, $i = 1, \ldots, r$ $\mathcal{N}_i$, $i = 1, \ldots, s$ filtrations

$$0 = \mathcal{E}_0 \subset \mathcal{E}_1 \subset \mathcal{E}_2 \subset \ldots \subset \mathcal{E}_r = \mathcal{E} \quad \text{and} \quad 0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \ldots \subset \mathcal{F}_s = \mathcal{F}$$

such that $\mathcal{E}_i/\mathcal{E}_{i-1} \cong \mathcal{L}_i$ and such that $\mathcal{F}_j/\mathcal{F}_{j-1} \cong \mathcal{N}_j$. Ordering pairs (i, j) lexicographically we obtain a filtration

$$0 \subset \ldots \subset \mathcal{E}_i \otimes \mathcal{F}_j + \mathcal{E}_{i-1} \otimes \mathcal{F} \subset \ldots \subset \mathcal{E} \otimes \mathcal{F}$$

with successive quotients

$$\mathcal{L}_1 \otimes \mathcal{N}_1, \mathcal{L}_1 \otimes \mathcal{N}_2, \ldots, \mathcal{L}_1 \otimes \mathcal{N}_s, \mathcal{L}_2 \otimes \mathcal{N}_1, \ldots, \mathcal{L}_r \otimes \mathcal{N}_s$$

By Lemma 02UJ we have

$$c(\mathcal{E}) = \prod(1 + x_i), \quad c(\mathcal{F}) = \prod(1 + y_j), \quad \text{and} \quad c(\mathcal{E} \otimes \mathcal{F}) = \prod(1 + x_i + y_j),$$

in $A^*(X)$. The result follows from a formal computation which we omit. □